

Forecasting One-Month-Ahead S&P 500 Returns

A Comparison of Macroeconomic Predictors and Regression Models

Section 1: Introduction

This project examines whether macroeconomic indicators can improve one-month-ahead forecasts of S&P 500 returns using monthly data from 2000 to 2025. The forecasting target is the following month's S&P 500 return.

An initial set of candidate predictors was collected from FRED, transformed using several methods, and adjusted for publication timing so that each forecast uses only information treated as available at the end of that month.

The predictors were assessed using stationarity tests, while correlation and variance inflation factor screening were conducted on the training set before a final set of six indicators was selected for modelling.

Forecasts were generated using a one-step-ahead forecasting procedure with a historical-mean benchmark, ordinary least squares, Ridge regression, and Lasso regression. Model performance was evaluated using RMSE, MAE, directional accuracy, and out-of-sample R^2 , and robustness was checked across several train-test splits.

Section 2: Data

Data Sources and Candidate Predictors

Ten macroeconomic indicators were collected as the initial predictor set, alongside the S&P 500 return series used as the forecasting target.

Variable	Source	FRED Series	Native Frequency
S&P 500 return	Yahoo Finance (<code>^GSPC</code>)	—	Daily → resampled monthly
CPI (inflation)	FRED	<code>CPIAUCSL</code>	Monthly
PPI	FRED	<code>PPIACO</code>	Monthly
M2 money supply	FRED	<code>M2SL</code>	Monthly
CapEx orders	FRED	<code>NEWORDER</code>	Monthly
Industrial production	FRED	<code>INDPRO</code>	Monthly
Unemployment rate	FRED	<code>UNRATE</code>	Monthly
Consumer sentiment	FRED	<code>UMCSENT</code>	Monthly
Federal funds rate	FRED	<code>FEDFUNDS</code>	Monthly
Term spread	FRED	<code>T10Y3M</code>	Daily → resampled monthly
Credit spread	FRED	<code>BAA</code> , <code>AAA</code>	Monthly

S&P 500 price and monthly return

This section downloads daily S&P 500 prices, smooths the closing price using a 5-day trailing average, and converts the series to monthly frequency using the final observation of each month. It then calculates monthly returns and creates the one-month-ahead return variable used as the forecasting target. Data was collected beginning in 1999 so that lagged variables and year-over-year percentage changes could be calculated without losing observations from the intended 2000–2025 sample.

Table showing head of the S&P 500 monthly dataset:

	<code>sp500_close_5day_avg</code>	<code>sp500_ret</code>	<code>sp500_ret_next</code>
Date			
1999-01-31	1254.894019	NaN	0.000894
1999-02-28	1256.016016	0.000894	0.030254
1999-03-31	1294.016016	0.030254	0.043536
1999-04-30	1350.352026	0.043536	-0.040389
1999-05-31	1295.812012	-0.040389	0.032032

Candidate Variable Downloads

Data for the ten candidate macroeconomic predictors were downloaded and aligned by date. The table provides a preview of the raw monthly data before any transformations or publication lags were applied.

date	1999-01-01	1999-02-01	1999-03-01	1999-04-01	1999-05-01
cpi	164.7000	164.700	164.8000	165.9000	166.0000
unemployment	4.3000	4.400	4.2000	4.3000	4.2000
consumer_sentiment	103.9000	108.100	105.7000	104.6000	106.8000
m2	4401.7000	4423.700	4430.1000	4459.9000	4487.9000
capex_orders	57160.0000	58576.000	59657.0000	58929.0000	59814.0000
indpro	87.4949	88.027	88.2267	88.4295	88.9886
fed_funds_rate	4.6300	4.760	4.8100	4.7400	4.7400
ppi	122.9000	122.300	122.6000	123.6000	124.7000
term_spread	NaN	0.220	0.6900	0.8300	NaN
baa_yield	7.2900	7.390	7.5300	7.4800	7.7200
aaa_yield	6.2400	6.400	6.6200	6.6400	6.9300

Predictor Transformations

The candidate variables were given transformations based on their economic interpretation. CPI, PPI, M2 money supply, capital expenditures, industrial production, and consumer sentiment were converted to year-over-year percentage growth rates. The unemployment rate, federal funds rate, term spread, and credit spread were first-differenced.

Isolated one-month gaps were forward-filled using the most recently available observation. This avoids using future values when replacing missing data and keeps the preprocessing consistent with the forecasting setup.

These transformations were only an initial preprocessing step. No formal stationarity testing had been conducted at this stage, so the specifications shown here were not necessarily the final transformations used in the forecasting models.

The table below provides a preview of the monthly data after the preliminary growth and first-difference transformations were applied.

date	2000-01-31	2000-02-29	2000-03-31	2000-04-30	2000-05-31
cpi	2.7930	3.2180	3.7621	3.0139	3.1325
ppi	4.3938	6.1325	6.6884	5.7443	5.5333
m2	6.0068	5.7644	6.2978	6.8724	6.0073
consumer_sentiment	7.7960	2.9602	1.3245	4.3977	3.6517
capex_orders	11.9227	-0.1349	6.6799	9.3078	6.5854
indpro	4.6210	4.3133	4.4473	4.8280	4.4515
unemployment	0.0000	0.1000	-0.1000	-0.2000	0.2000
fed_funds_rate	0.1500	0.2800	0.1200	0.1700	0.2500
term_spread	0.2434	-0.3695	-0.3999	-0.2237	0.2770
credit_spread	-0.0900	0.0600	0.0800	0.0700	0.1500

Merging Dataframes

The transformed macroeconomic indicators were merged with the S&P 500 dataset using their shared monthly date index. Rows containing missing values were then removed to create a complete dataset for the subsequent variable-selection and forecasting analysis.

The table below previews the initial merged dataset, including the transformed predictors, the five-day average S&P 500 closing price, the monthly return, and the one-month-ahead return used as the forecasting target.

	2000-01-31	2000-02-29	2000-03-31	2000-04-30	2000-05-31
cpi	2.7930	3.2180	3.7621	3.0139	3.1325
ppi	4.3938	6.1325	6.6884	5.7443	5.5333
m2	6.0068	5.7644	6.2978	6.8724	6.0073
consumer_sentiment	7.7960	2.9602	1.3245	4.3977	3.6517
capex_orders	11.9227	-0.1349	6.6799	9.3078	6.5854
indpro	4.6210	4.3133	4.4473	4.8280	4.4515
unemployment	0.0000	0.1000	-0.1000	-0.2000	0.2000
fed_funds_rate	0.1500	0.2800	0.1200	0.1700	0.2500
term_spread	0.2434	-0.3695	-0.3999	-0.2237	0.2770
credit_spread	-0.0900	0.0600	0.0800	0.0700	0.1500
sp500_close_5day_avg	1393.4600	1352.3900	1505.3220	1457.1280	1400.3280
sp500_ret	-0.0471	-0.0295	0.1131	-0.0320	-0.0390
sp500_ret_next	-0.0295	0.1131	-0.0320	-0.0390	0.0366

Publication-Lag Treatment of Candidate Predictors

The publication lags below are separate from the one-month-ahead forecasting setup. The target variable is already defined as the S&P 500 return in month (t+1), while each row represents the information available by the end of month (t). These additional shifts account for whether a month-(t) observation was publicly available at the forecast date.

The timing adjustments are applied to all candidate predictors before calculating correlations, VIF values, and selecting the final predictor set.

Variable	Data timing	Publication lag applied		Reason
CPI	The month-(t) observation is released during month (t+1)	1 month		The month-(t) value is not known by the end of month (t)
PPI	The month-(t) observation is released during month (t+1)	1 month		The month-(t) value is not known by the end of month (t)
M2 money supply	The month-(t) observation is released during month (t+1)	1 month		The month-(t) value is not known by the end of month (t)
CapEx orders	The month-(t) observation is released during month (t+1)	1 month		The month-(t) value is not known by the end of month (t)
Industrial production	The month-(t) observation is released during month (t+1)	1 month		The month-(t) value is not known by the end of month (t)
Unemployment rate	The month-(t) observation is released during month (t+1)	1 month		The month-(t) value is not known by the end of month (t)
Consumer sentiment	Preliminary and final readings are released during the reference month	1 month		A conservative lag avoids relying on preliminary or revised within-month information
Federal funds rate	The underlying daily effective federal funds rate is observed during month (t)	None		The month-(t) value is known by the end of month (t)
Term spread	Constructed from daily Treasury yields available during month (t)	None		The monthly average through the end of month (t) is known at the forecast date
Credit spread	Constructed from monthly Baa and Aaa corporate-yield series	1 month		The monthly observation is treated conservatively as unavailable until the following month

Therefore, CPI, PPI, M2, CapEx orders, industrial production, unemployment, consumer sentiment, and the credit spread are shifted by one month. The federal funds rate and term spread are left unshifted because their underlying market information is available by month-end. The resulting aligned predictors are then used to examine correlations with the S&P 500 return in the following month.

	2000-02-29	2000-03-31	2000-04-30	2000-05-31	2000-06-30
cpi	2.7930	3.2180	3.7621	3.0139	3.1325
ppi	4.3938	6.1325	6.6884	5.7443	5.5333
m2	6.0068	5.7644	6.2978	6.8724	6.0073
consumer_sentiment	7.7960	2.9602	1.3245	4.3977	3.6517
capex_orders	11.9227	-0.1349	6.6799	9.3078	6.5854
indpro	4.6210	4.3133	4.4473	4.8280	4.4515
unemployment	0.0000	0.1000	-0.1000	-0.2000	0.2000
fed_funds_rate	0.2800	0.1200	0.1700	0.2500	0.2600
term_spread	-0.3695	-0.3999	-0.2237	0.2770	-0.2105
credit_spread	-0.0900	0.0600	0.0800	0.0700	0.1500
sp500_close_5day_avg	1352.3900	1505.3220	1457.1280	1400.3280	1451.5340
sp500_ret	-0.0295	0.1131	-0.0320	-0.0390	0.0366
sp500_ret_next	0.1131	-0.0320	-0.0390	0.0366	-0.0042

The table below shows the predictor values available at each month-end after accounting for publication timing. These values are used to forecast the following month's S&P 500 return.

Stationarity and Predictor Screening

The candidate predictors were tested for stationarity using the Augmented Dickey–Fuller (ADF) and KPSS tests. The table below reports the p-values and stationarity conclusions from both tests for each variable.

	variable	ADF p-value	KPSS p-value	ADF conclusion	KPSS conclusion
0	cpi	2.647465e-02	0.1	Stationary	Stationary
1	ppi	1.134225e-04	0.1	Stationary	Stationary
2	m2	6.496385e-03	0.1	Stationary	Stationary
3	capex_orders	5.390194e-04	0.1	Stationary	Stationary
4	indpro	4.724559e-03	0.1	Stationary	Stationary
5	unemployment	1.919731e-18	0.1	Stationary	Stationary
6	consumer_sentiment	3.058016e-03	0.1	Stationary	Stationary
7	fed_funds_rate	1.067492e-03	0.1	Stationary	Stationary
8	term_spread	1.360527e-13	0.1	Stationary	Stationary
9	credit_spread	1.919805e-19	0.1	Stationary	Stationary

All predictor variables are classified as stationary under both the ADF and KPSS tests. These transformed series are therefore retained for the subsequent predictor-screening and forecasting analysis.

Final Variable Transformations

The table below summarizes the transformation applied to each predictor. It also provides a brief interpretation of what each transformed variable represents in the forecasting dataset.

Variable	Transformation used	Interpretation
CPI	Year-over-year percentage growth	Annual CPI inflation rate
PPI	Year-over-year percentage growth	Annual producer-price inflation rate
M2 money supply	Year-over-year percentage growth	Annual growth in the money supply
CapEx orders	Year-over-year percentage growth	Annual growth in new capital-goods orders
Industrial production	Year-over-year percentage growth	Annual growth in industrial production
Consumer sentiment	Year-over-year percentage growth	Annual growth in the consumer sentiment index
Unemployment rate	First difference	Monthly change in unemployment rate
Federal funds rate	First difference	Monthly change in effective federal funds rate
Term spread	First difference	Monthly change in the 10-year minus 3-month Treasury spread

Variable	Transformation used	Interpretation
Credit spread	First difference	Monthly change in the difference between Baa and Aaa corporate bond yields

Section 3: Predictor Screening and Selection

Correlation Screening

The correlations between each transformed predictor and the one-month-ahead S&P 500 return were calculated using only the initial training sample. Restricting the screening to the training sample prevents the testing period from influencing variable selection and provides a more credible measure of out-of-sample performance. The results are ordered by absolute correlation so that the predictors with the strongest individual relationships with next-month returns appear first.

```

cpi          -0.227385
credit_spread -0.207904
ppi          -0.177276
fed_funds_rate 0.171603
unemployment  -0.152950
consumer_sentiment 0.136984
term_spread  -0.063341
capex_orders  0.025722
indpro        0.025237
m2            0.019369
dtype: float64

```

Overall, the correlations are weak. CPI and PPI have the largest absolute correlations, while consumer sentiment and capital expenditures have relationships close to zero. Correlation is used only as an initial screening tool and does not by itself establish that a variable will improve out-of-sample forecasts.

VIF Analysis

Variance inflation factors were calculated for the full set of candidate predictors to identify variables containing highly overlapping information. A high VIF indicates that a predictor is strongly related to other predictors, which can make regression coefficients unstable.

	variable	VIF
0	ppi	10.241449
1	cpi	6.085652
2	capex_orders	5.445323
3	indpro	3.793871
4	consumer_sentiment	2.206676
5	m2	2.082070
6	fed_funds_rate	1.896380
7	unemployment	1.618743
8	term_spread	1.446829
9	credit_spread	1.260863

The final predictor set consists of CPI, industrial production, the term spread, the credit spread, the unemployment rate, and the federal funds rate. Predictor selection combined the correlation and VIF results with economic intuition about inflation, economic activity, monetary policy, and financial conditions.

	variable	VIF
0	indpro	1.923581
1	fed_funds_rate	1.652347
2	unemployment	1.494460
3	term_spread	1.434022
4	cpi	1.393221
5	credit_spread	1.202634

The VIF values were then recalculated for the final predictor set. All six values are low, indicating that the retained predictors do not create a meaningful multicollinearity problem.

Final Predictor Set

The final modelling dataset contains the six predictors used in the forecasting models: CPI, industrial production, the term spread, the credit spread, the unemployment rate, and the federal funds rate. These variables are combined with the one-month-ahead S&P 500 return, which serves as the forecasting target.

The table below previews the completed dataset that will be used to estimate and evaluate the forecasting models.

	cpi	fed_funds_rate	indpro	term_spread	unemployment	credit_spread	sp500_ret_next
2000-02-29	2.792957	0.28	4.620955	-0.369500	0.0	-0.09	0.113083
2000-03-31	3.217972	0.12	4.313336	-0.399891	0.1	0.06	-0.032016
2000-04-30	3.762136	0.17	4.447293	-0.223661	-0.1	0.08	-0.038981
2000-05-31	3.013864	0.25	4.828027	0.276962	-0.2	0.07	0.036567
2000-06-30	3.132530	0.26	4.451469	-0.210455	0.2	0.15	-0.004194

Section 4: Summary Statistics

Descriptive Statistics Table

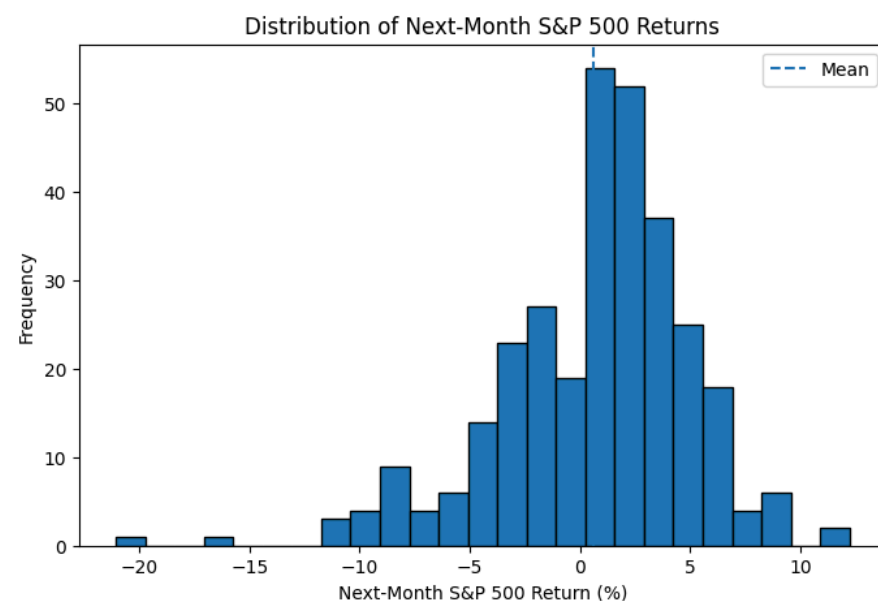
The table below summarizes the final predictors and the one-month-ahead S&P 500 return using the number of observations, mean, standard deviation, minimum, quartiles, and maximum. These statistics provide an initial comparison of each variable's typical value, variability, and overall range before the forecasting models are estimated.

	N	Mean	Std. Dev.	Minimum	25th Percentile	Median	75th Percentile	Maximum
CPI Inflation (YoY %)	309.0	2.5824	1.7382	-1.9588	1.6406	2.3321	3.3159	8.9794
Change in Federal Funds Rate (Percentage Points)	309.0	-0.0044	0.1828	-0.9600	-0.0100	0.0000	0.0200	0.7000
Industrial Production Growth (YoY %)	309.0	0.5672	4.2823	-17.3183	-0.9971	1.6574	2.9076	16.5521
Change in Term Spread (Percentage Points)	309.0	-0.0035	0.2420	-1.0068	-0.1546	-0.0165	0.1199	0.8250
Change in Unemployment Rate (Percentage Points)	309.0	0.0013	0.6475	-2.2000	-0.1000	0.0000	0.1000	10.4000
Change in Credit Spread (Percentage Points)	309.0	-0.0001	0.1174	-0.6300	-0.0500	-0.0100	0.0400	0.9400
Next-Month S&P 500 Return (Decimal)	309.0	0.0062	0.0436	-0.2106	-0.0183	0.0122	0.0335	0.1223

The variables differ noticeably in scale and variability. CPI inflation and the unemployment rate show particularly wide ranges, while changes in the term spread are centered close to zero. The average next-month S&P 500 return is positive, but its minimum and maximum show that monthly returns can vary substantially.

Distribution of Next-Month S&P 500 Returns

The histogram below shows the distribution of the one-month-ahead S&P 500 return used as the forecasting target. The dashed vertical line marks the average return and provides a reference point for the historical-mean benchmark used later in the analysis.



The distribution of next-month S&P 500 returns is centered slightly above zero, with most observations falling between roughly -5% and 5%. The distribution is negatively skewed because of a small number of large negative returns, including one extreme observation near -20%. Positive returns occur more frequently than negative returns, which helps explain why the historical-mean benchmark achieves relatively high directional accuracy despite making fairly simple forecasts.

Section 5: Model Estimation and Evaluation

The forecasting models are evaluated using 60/40, 70/30, and 80/20 chronological train-test splits. OLS, Ridge, and Lasso are re-estimated each month using the most recent 120 months, while the historical-mean benchmark uses all previously observed returns. Performance is compared using RMSE, MAE, directional accuracy, and out-of-sample R^2 across the three testing periods.

60/40 Train-Test Split

	RMSE	MAE	Directional Accuracy	Out-of-Sample R^2
Model				
Historical-Mean Benchmark	0.0434	0.0334	0.7016	0.0000
OLS	0.0610	0.0371	0.6210	-0.9739
Ridge	0.0436	0.0328	0.7016	-0.0088
Lasso	0.0438	0.0328	0.6774	-0.0175

70/30 Train-Test Split

	RMSE	MAE	Directional Accuracy	Out-of-Sample R^2
Model				
Historical-Mean Benchmark	0.0460	0.0361	0.7097	0.0000
OLS	0.0673	0.0405	0.6022	-1.1349
Ridge	0.0459	0.0349	0.7097	0.0040
Lasso	0.0462	0.0351	0.6774	-0.0075

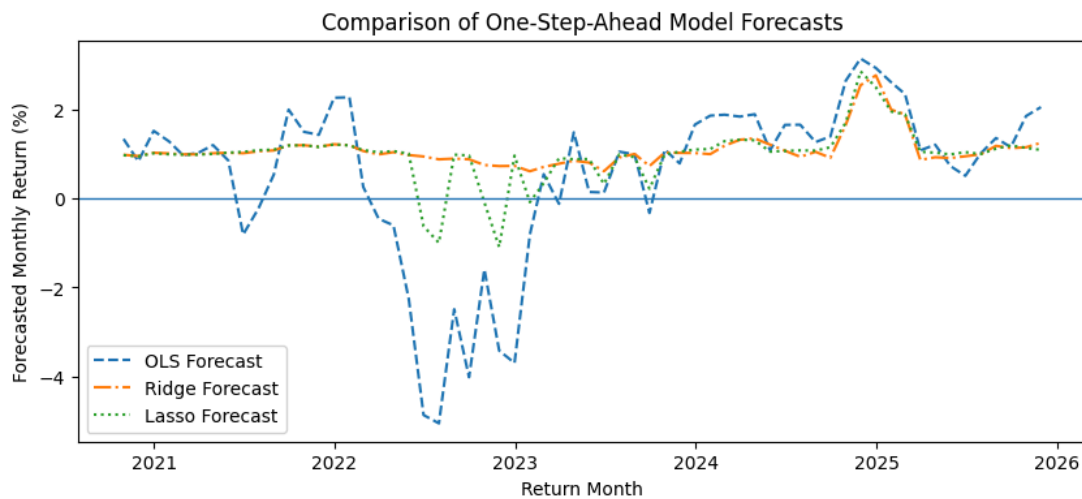
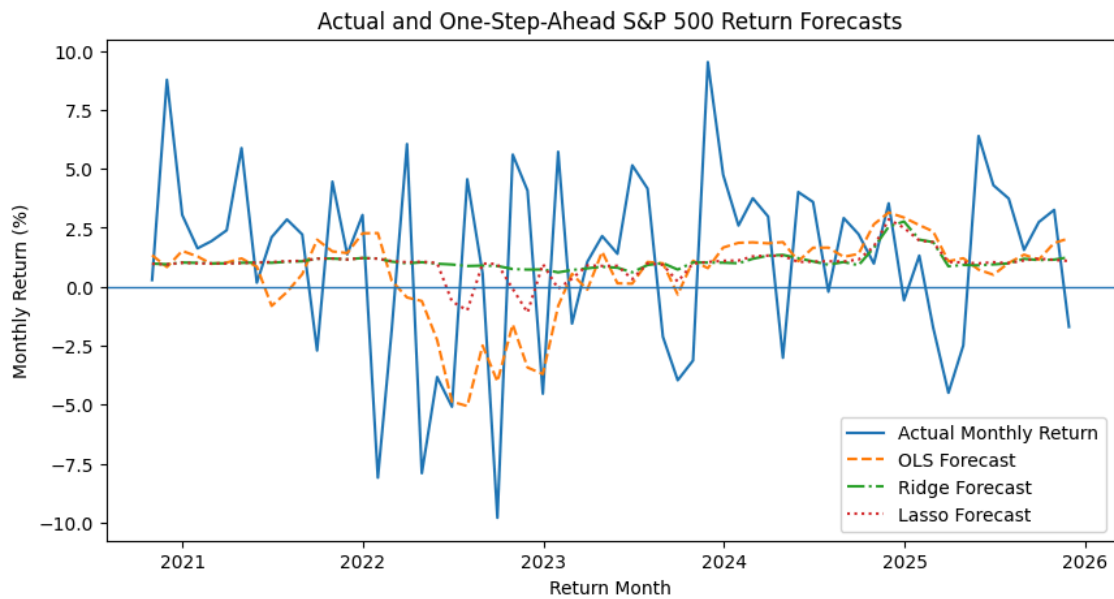
80/20 Train-Test Split

	RMSE	MAE	Directional Accuracy	Out-of-Sample R^2
Model				
Historical-Mean Benchmark	0.0396	0.0326	0.6935	0.0000
OLS	0.0403	0.0317	0.6452	-0.0353
Ridge	0.0394	0.0316	0.6935	0.0146
Lasso	0.0399	0.0319	0.6452	-0.0153

The models use 120-month rolling training windows and are evaluated across three forecast starting points. Ridge achieves approximately 1.5% out-of-sample R^2 in the 80/20 split, but does not consistently outperform the historical-mean benchmark across all three splits.

Visual Comparison of Model Forecasts

The first plot compares the actual one-month-ahead S&P 500 returns with the forecasts from OLS, Ridge, and Lasso over the test period. The second plot removes the actual return series and focuses only on the model forecasts, making it easier to compare how the three models differ in their predicted movements.



The first figure shows that all three forecasting models produce much smoother predictions than the realized monthly S&P 500 returns. None of the models captures the large positive and negative movements observed during the test period, which is consistent with their weak out-of-sample performance.

The second figure highlights the differences across the model forecasts. OLS produces the most variable predictions and reacts most strongly to changes in the selected predictors. Ridge reduces some of this variation through coefficient shrinkage, while Lasso produces slightly smoother forecasts. Despite these differences, all three models fail to match the volatility of actual monthly returns, although Ridge slightly outperforms the historical-mean benchmark.

Section 6: Discussion and Conclusion

The results differed across the models. In the 80/20 split, Ridge produced slightly lower forecast errors than the historical-mean benchmark, achieving approximately 1.5% out-of-sample R^2 . This was a modest improvement, while the other regression models did not show the same result. The forecast graphs also showed that none of the models captured the larger month-to-month movements in actual returns particularly well.

The robustness checks using alternative train-test splits showed that performance depended on the forecast starting point. Although Ridge slightly outperformed the benchmark in the 70/30 and 80/20 splits, it underperformed in the 60/40 split.

Overall, the selected macroeconomic indicators provided limited predictive value for monthly S&P 500 returns. The Ridge result shows that a small improvement over the historical mean is possible, but the results do not establish a consistent forecasting advantage.